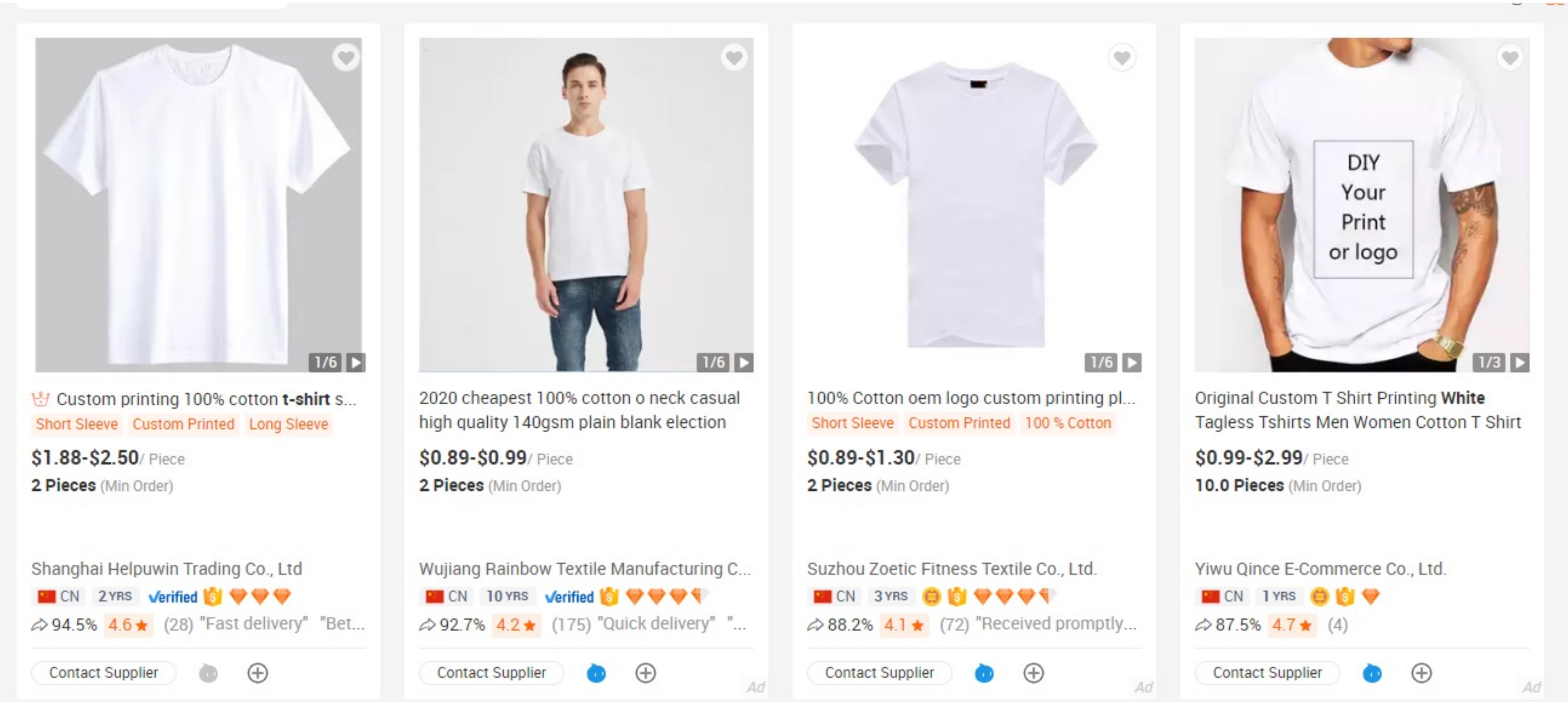
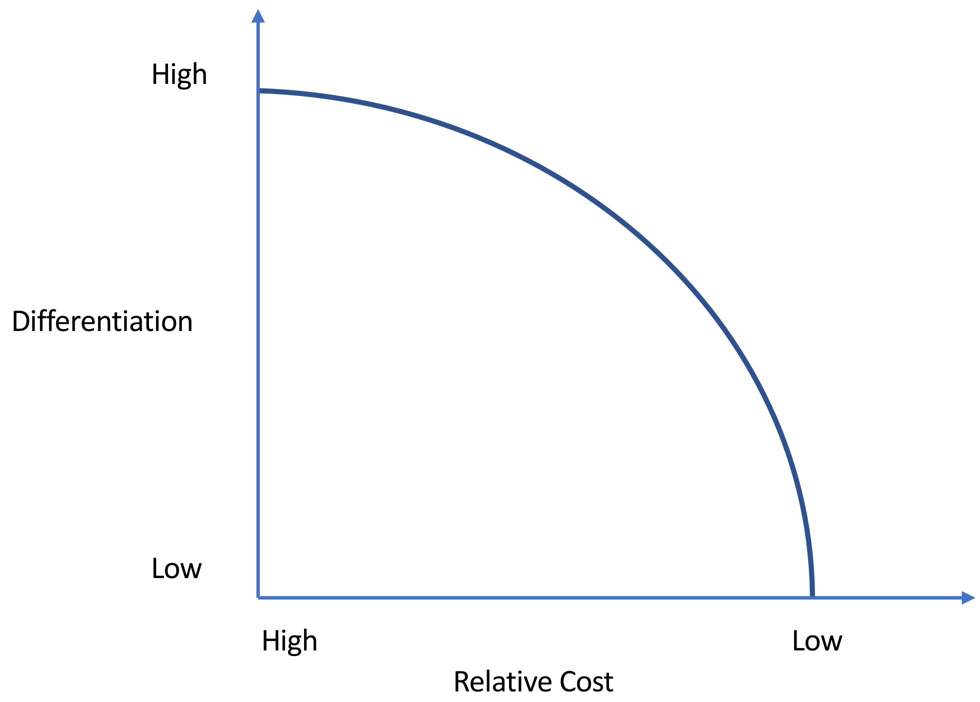
Basic Microeconomics

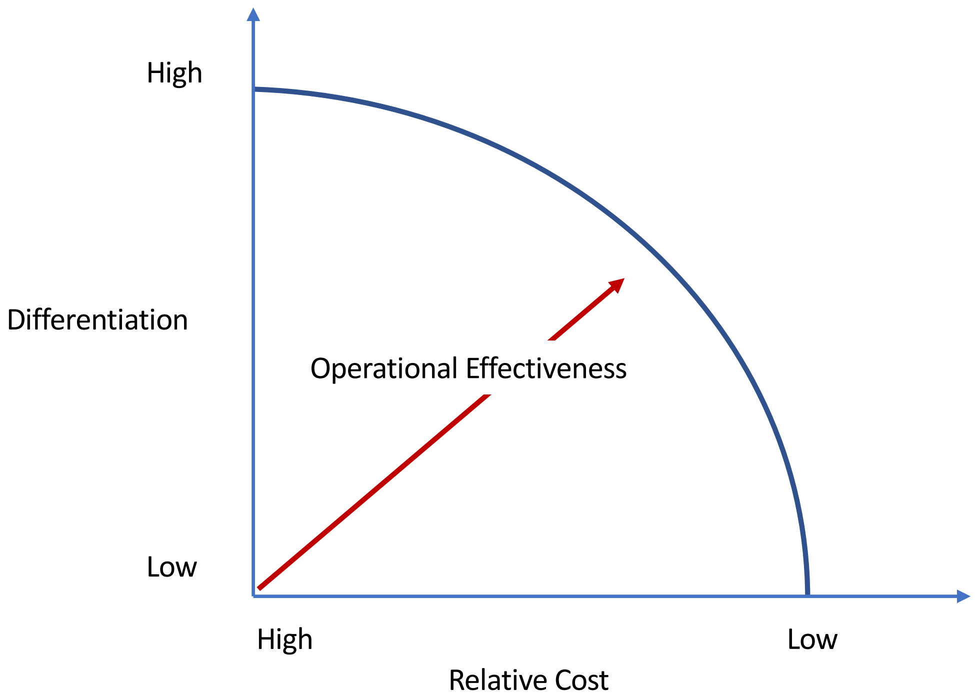


- How much does it cost to produce a white T-shirt?
- How much would you price your white T-shirt?
- What happens in the long run?

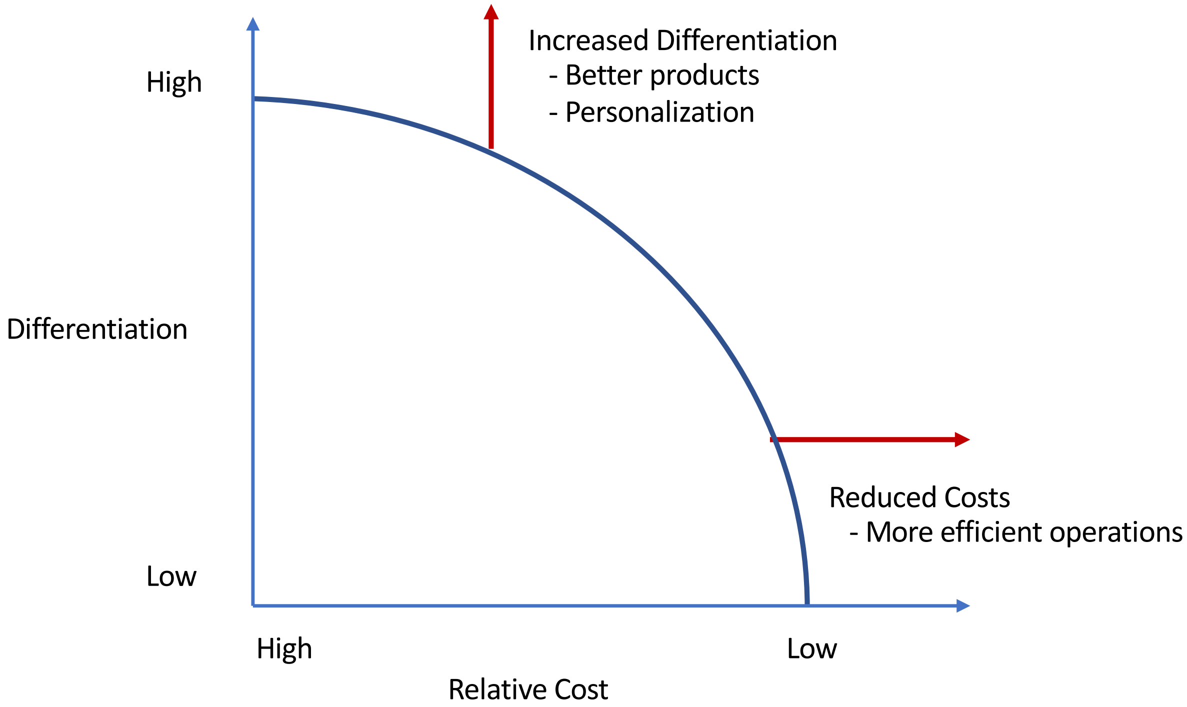
Differentiation and Costs



Analytics, Differentiation, and Costs (Cont'd)

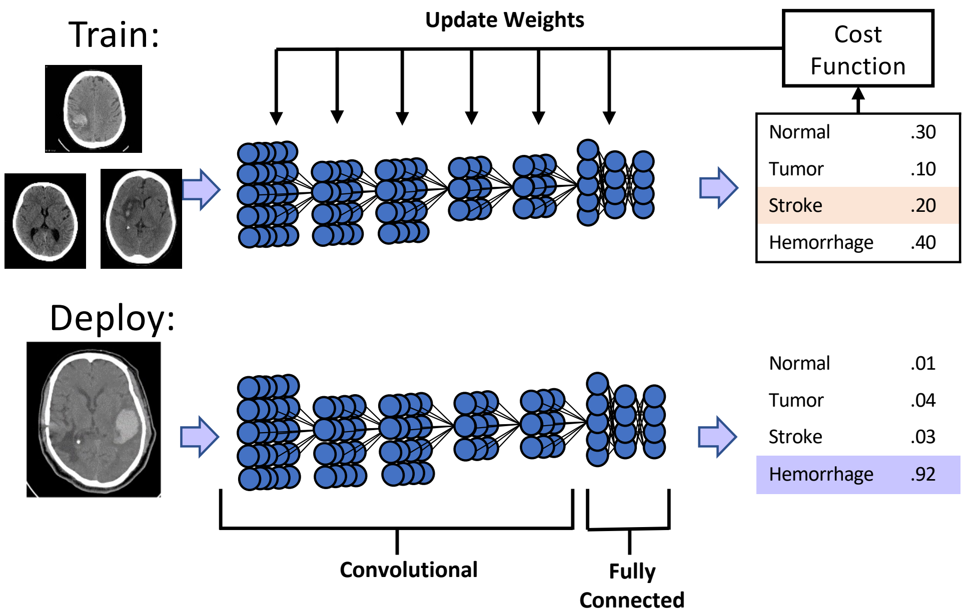


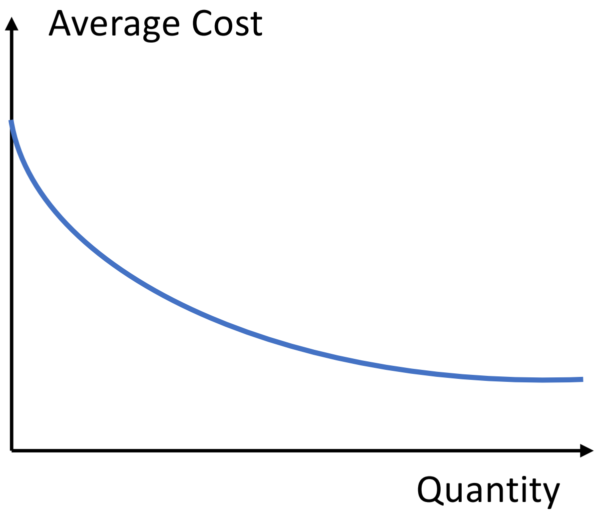
Analytics, Differentiation, and Costs (Cont’d)



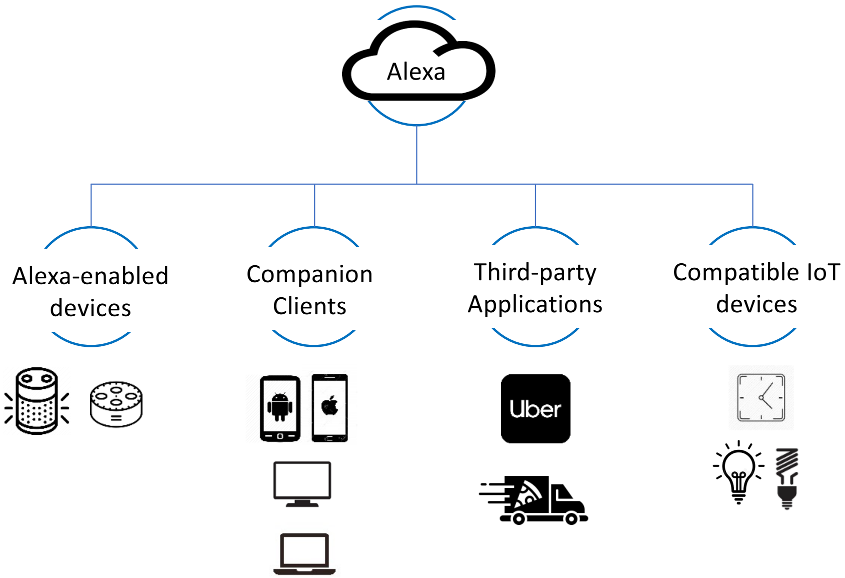
Analytics and Economies of Scale

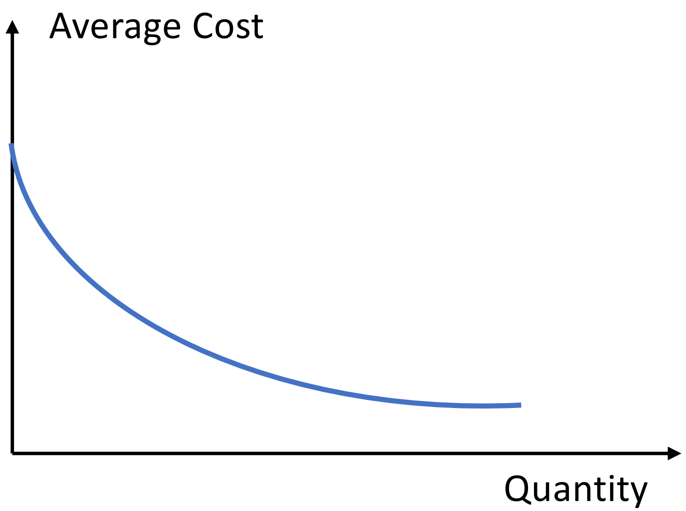
Convolutional Neural Networks



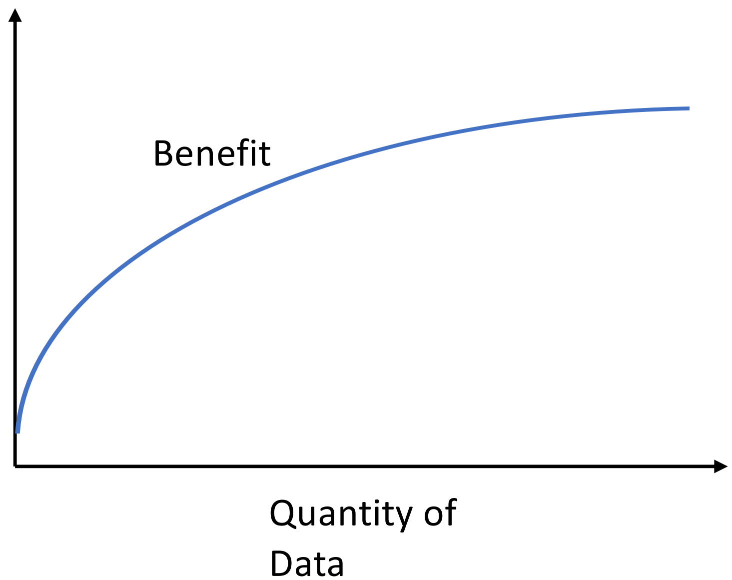


Analytics and Economies of Scope

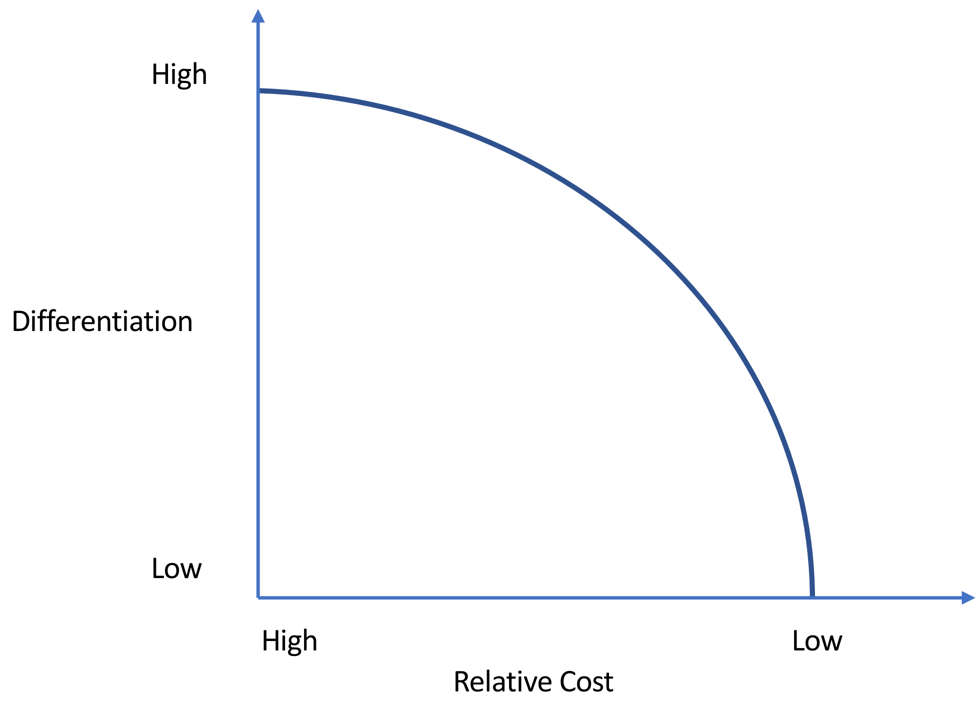




Cumulative Advantage?



Differentiation and Costs

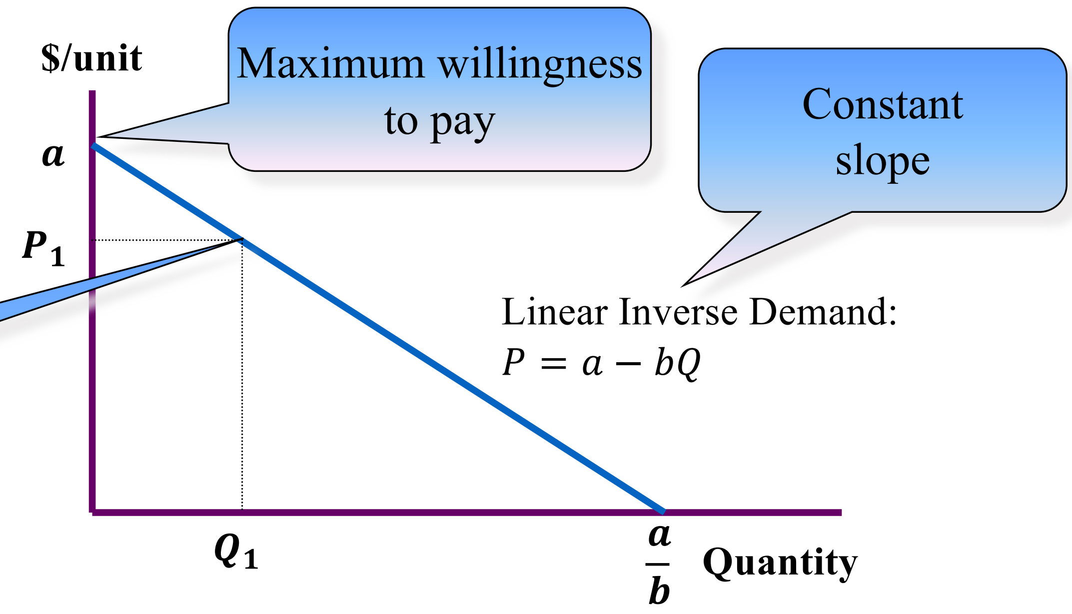


Efficiency and market performance

- Contrast two polar cases
  - perfect competition
  - monopoly
- What is *efficiency?*
  - no reallocation of the available resources makes one economic agent better off without making some other economic agent worse off
  - example: given an initial distribution of food aid will trade between recipients improve efficiency?

Profit maximization: the basics

- Focus on profit maximizing behavior of firms
- Take as given the **market demand curve**
- Importance of:
  - Time
  - Short-run vs. long-run
  - Willingness-to-pay



The first order condition: MR = MC

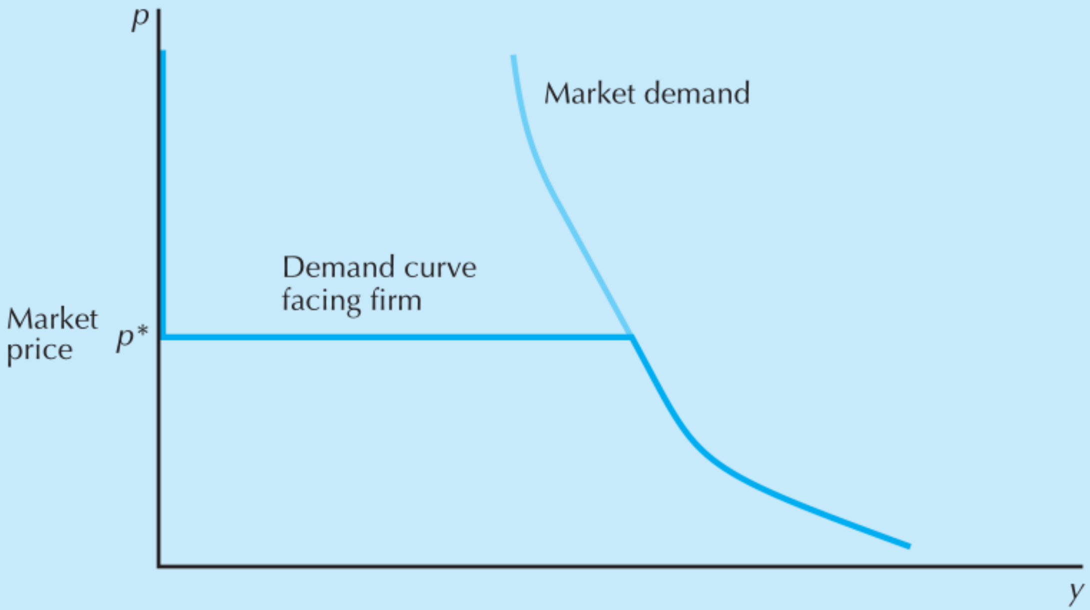
- Profit is $\pi(q) = R(q) - C(q)$
- Profit maximization: $\frac{d\pi}{dq} = 0$
- This implies $\frac{dR(q)}{dq} - \frac{dC(q)}{dq} = 0,$
  where $\frac{dR(q)}{dq}$ = marginal revenue (MR) and $\frac{dC(q)}{dq}$ = marginal cost (MC)
- So, profit maximization implies $MR \ = \ MC$

What could go wrong?

Perfect competition

- Firms and consumers are ***price-takers***
- Classic example: an identical product and many small firms
- Can you give an example with 3 – 4 firms?
- Firm can sell as much as it likes at the ruling market price
  - do not need many firms
  - do need the idea that firms ***believe*** that their actions will not affect the market price
- Therefore, **marginal revenue equals price**
- To maximize profit a firm *of any type* must equate marginal revenue with marginal cost
- So, in perfect competition price equals marginal cost: $p = MC$

Perfect competition (Cont'd)



The demand curve facing a competitive firm. The firm's demand is horizontal at the market price. At higher prices, the firm sells nothing, and below the market price it faces the entire market demand curve.

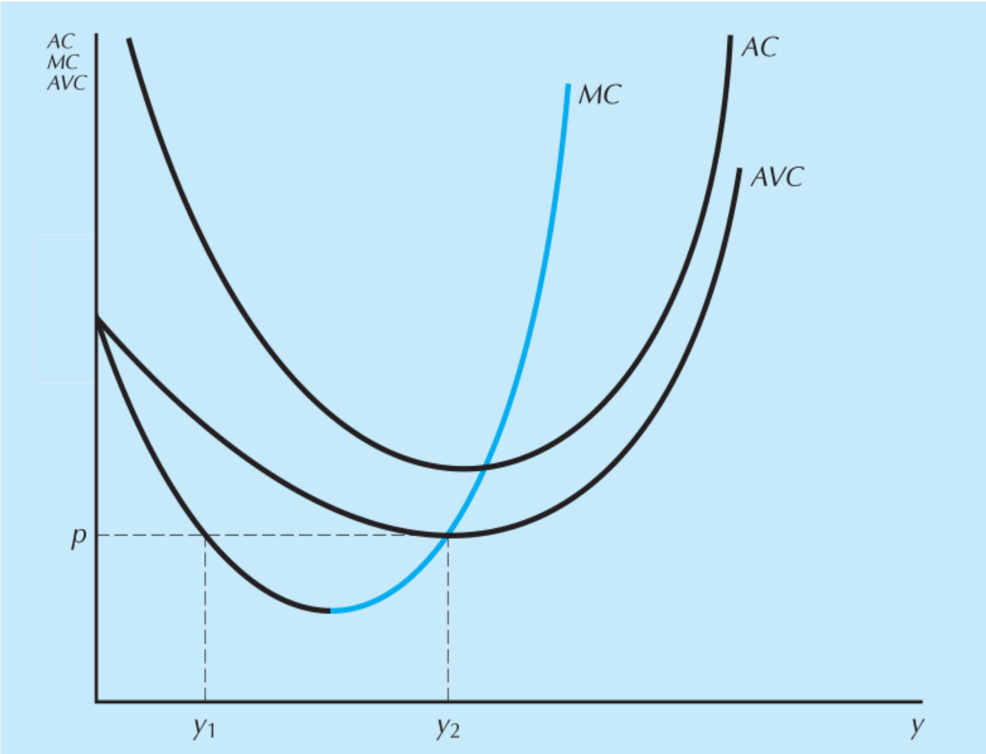
- The **market demand curve** measures the relationship between the market price and the total amount of output sold.
  - Depends on consumers' behavior

- The **demand curve facing a firm** measures the relationship between the market price and the output of *that particular firm.*
  - Depends not only on consumers' behavior
  - Also depends on the behavior of other firms
  - Usual justification for competitive model: many small firms
  - Question: Is it possible to have a competitive demand curve with only 2 firms?

The supply decision of a competitive firm

- $\max_{q} \pi(q) = \max_{q} Pq - c(q)$
- Remember: $MR = MC$
- At optimal level of output, firm must be producing where $P = MC(q)$
- The marginal cost is the **firm's supply curve**

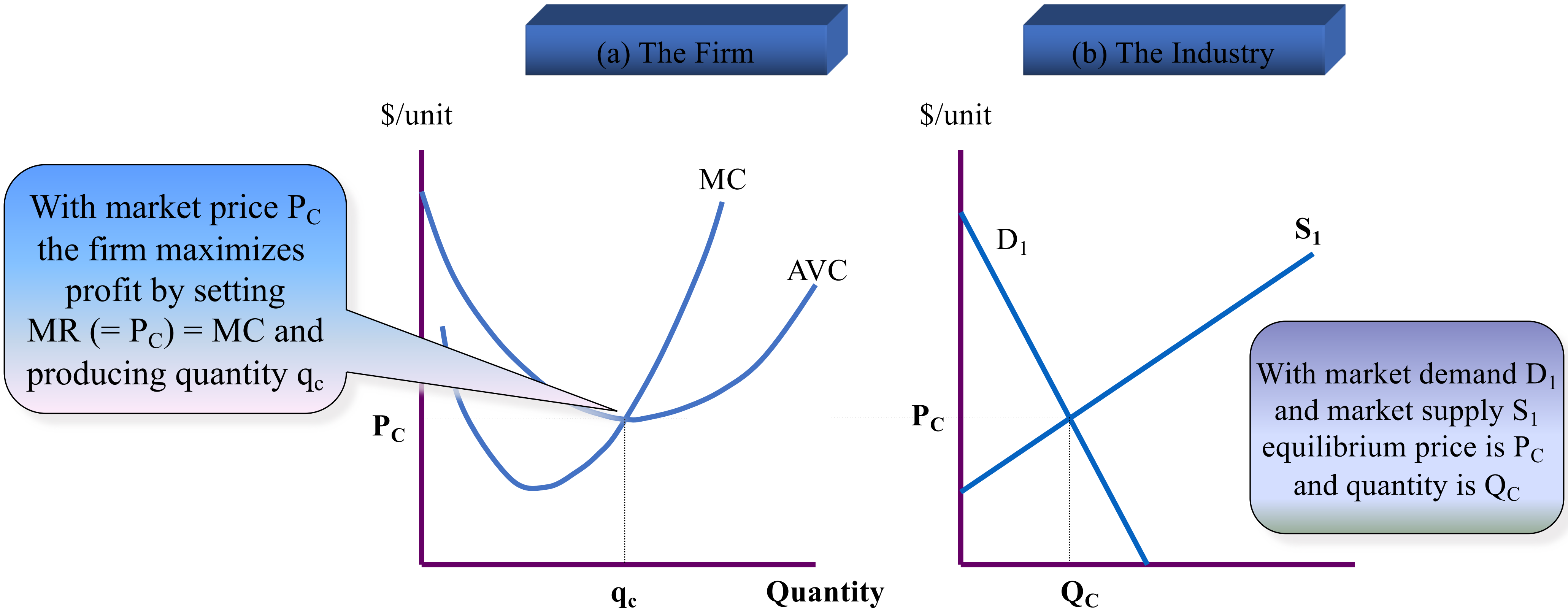
Necessary but not sufficient



MC = marginal cost, AVC = average variable cost,
AC = average cost

- There are two levels of output where $p = MC$
- The profit-maximizing quantity supplied can only lie on the upward-sloping part of the MC curve
- Supply curve is upward sloping

Perfect competition: an illustration

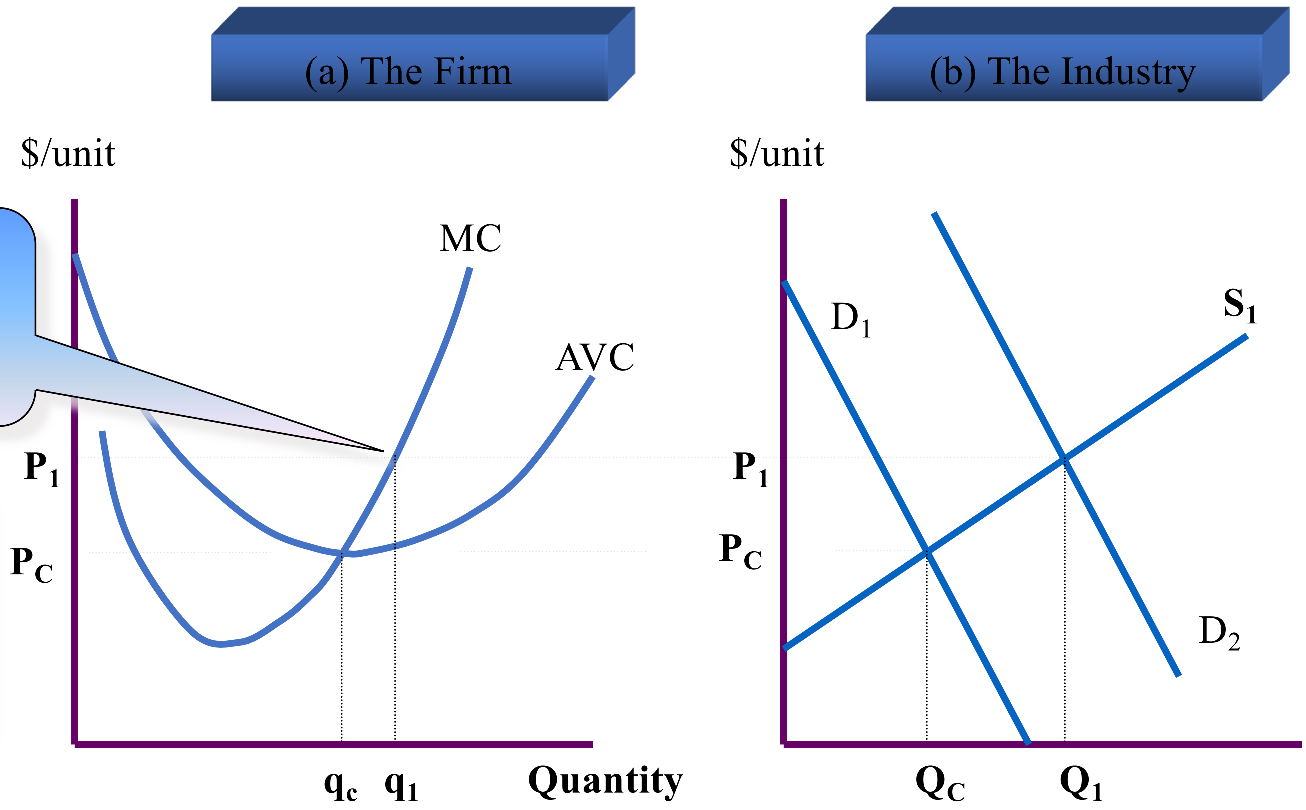


Perfect competition: an illustration

Now assume that demand increases to D_2

Existing firms maximize profits by increasing output to q_1

With market demand D_2 and market supply S_1 equilibrium price is P_1 and quantity is Q_1



Perfect competition: an illustration

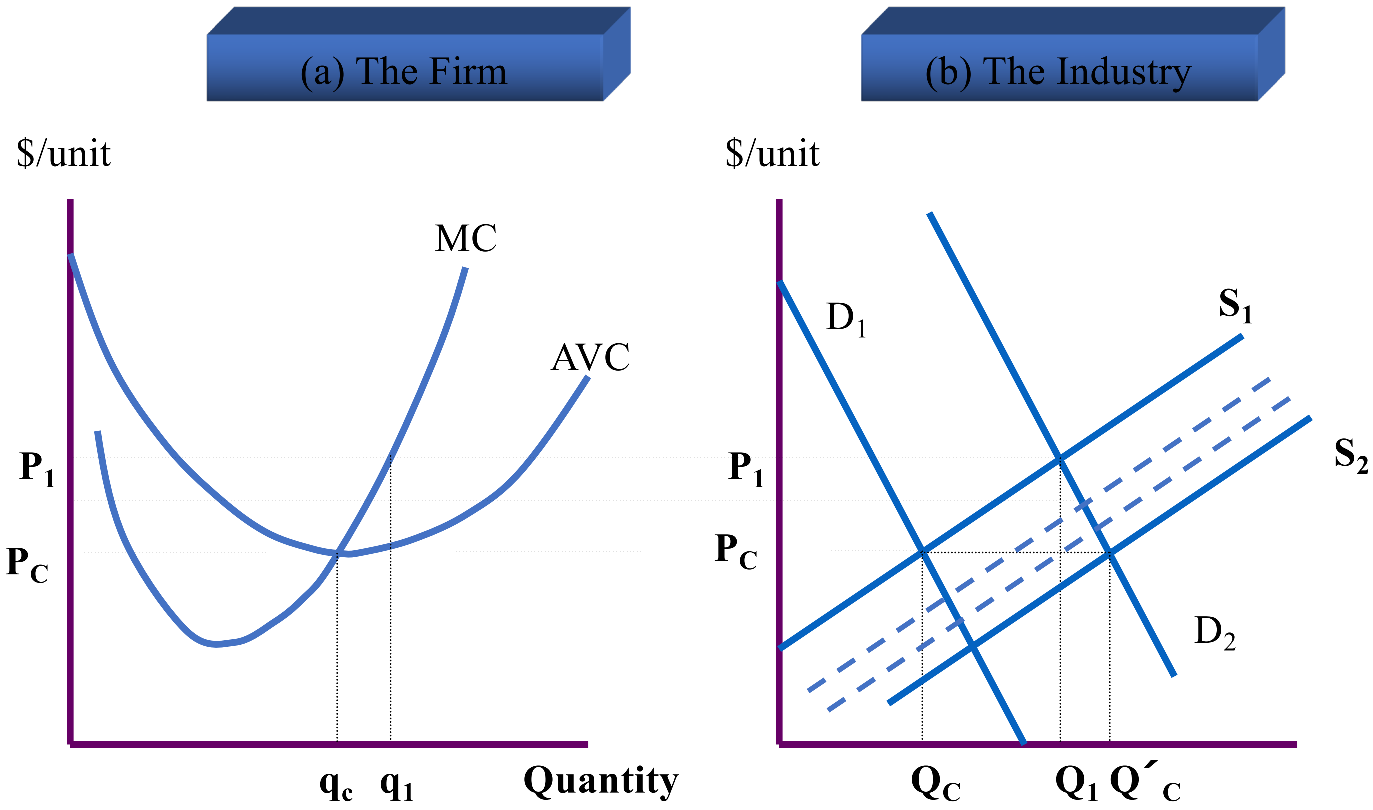
Excess profits induce new firms to enter the market

The supply curve moves to the right

Price falls

Entry continues while profits exist

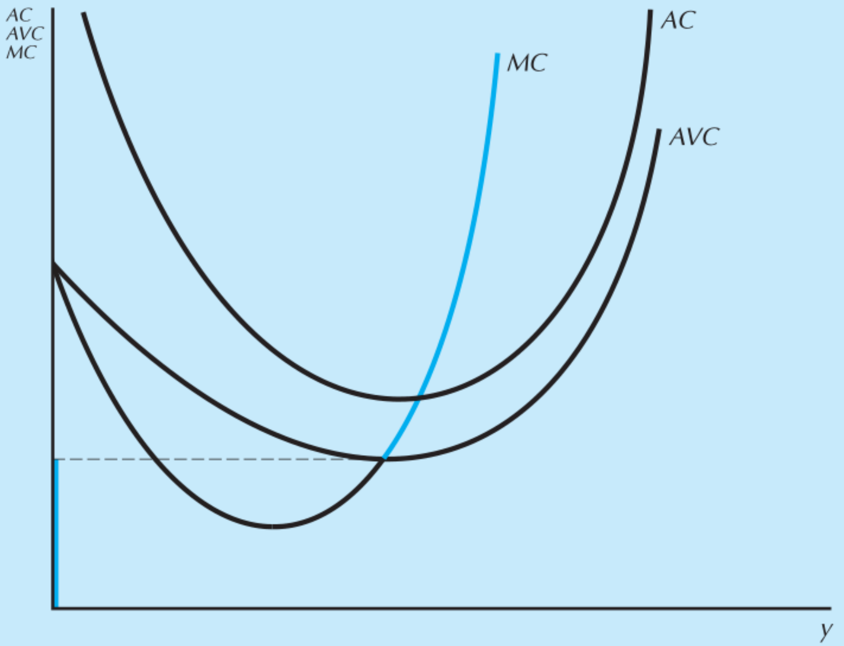
Long-run equilibrium is restored at price P_C and supply curve S_2



Costs

- **Total Costs** of firm $c(q) = c_v(q) + F$
- **Average cost function** measures the cost per unit of output
$$AC(q) = \frac{c(q)}{q} = \frac{c_v(q)}{q} + \frac{F}{q} = AVC(q) + AFC(q)$$
- In the short run, a firm producing zero output still has to pay F
- Firm is better going out of business (producing 0) when
$$-F > pq - c_v(q) - F$$
- **Shutdown condition:**
$$AVC(q) = \frac{c_v(q)}{q} > p$$

Back to the firm's supply curve



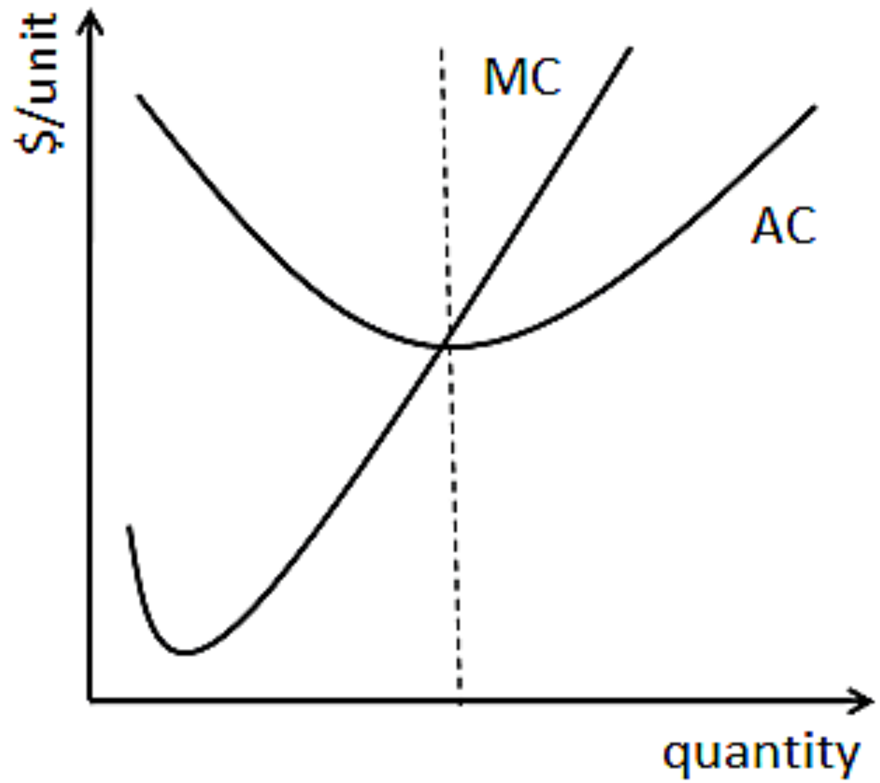
Average variable cost and supply. The supply curve is the upward-sloping part of the marginal cost curve that lies above the average variable cost curve. The firm will not operate on those points on the marginal cost curve below the average cost curve since it could have greater profits (less losses) by shutting down.

Average cost and marginal cost

- Average cost $= AC(q) =$ total cost / quantity $= \frac{c(q)}{q}$
- Marginal cost $= MC(q) = c(q+1) - c(q)$ or

$$MC(q) = \frac{c(q+\Delta q) - c(q)}{\Delta q} = \frac{\Delta c(q)}{\Delta q} = c'(q)$$

- So, we have $AC'(q) = \frac{c'(q)}{q} - \frac{c(q)}{q^2} = \frac{MC(q) - AC(q)}{q}$
- $AC'(q) > 0\ if\ MC(q) > AC(q), AC'(q) < 0\ if\ MC(q) < AC(q)$



MC < AC
AC decreasing

MC > AC
AC increasing

The short-run industrial supply curve

- Let $S_i(p)$ be the supply curve of firm i
- The **industry supply curve**, or the **market supply curve**:

$$S(p) = \sum_{i=1}^{n} S_i(p)$$

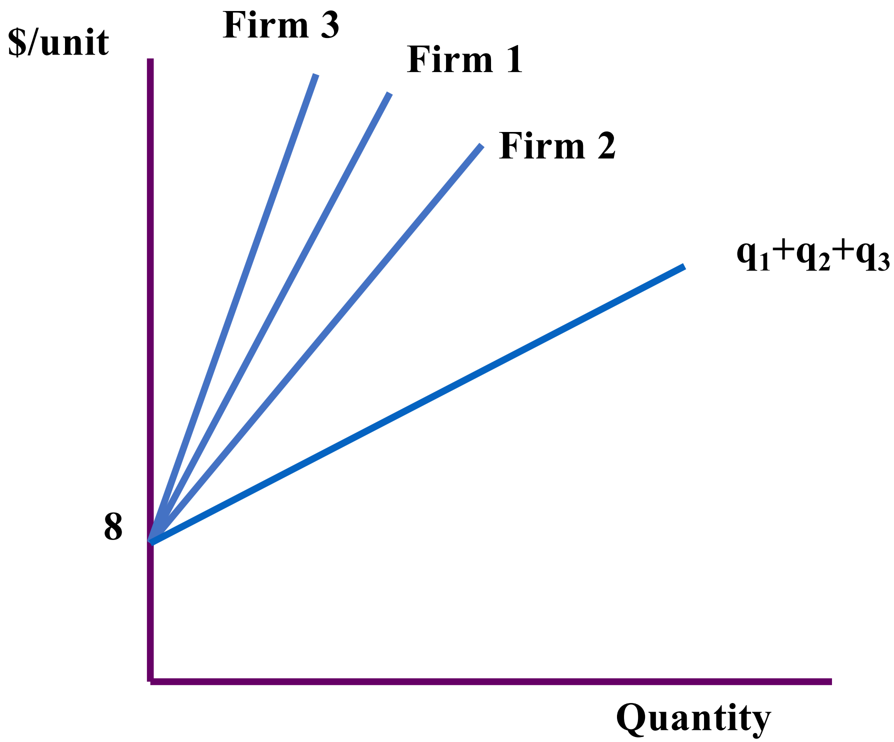
- Geometrically, *horizontal* sum of supply curves (*MC* curves)

Example: 3 firms

Firm 1: $MC = 4q + 8$

Firm 2: $MC = 2q + 8$

Firm 3: $MC = 6q + 8$



Example: 3 firms (Cont'd)

$$\text{Firm 1: } q_1 = \frac{MC}{4} - 2$$

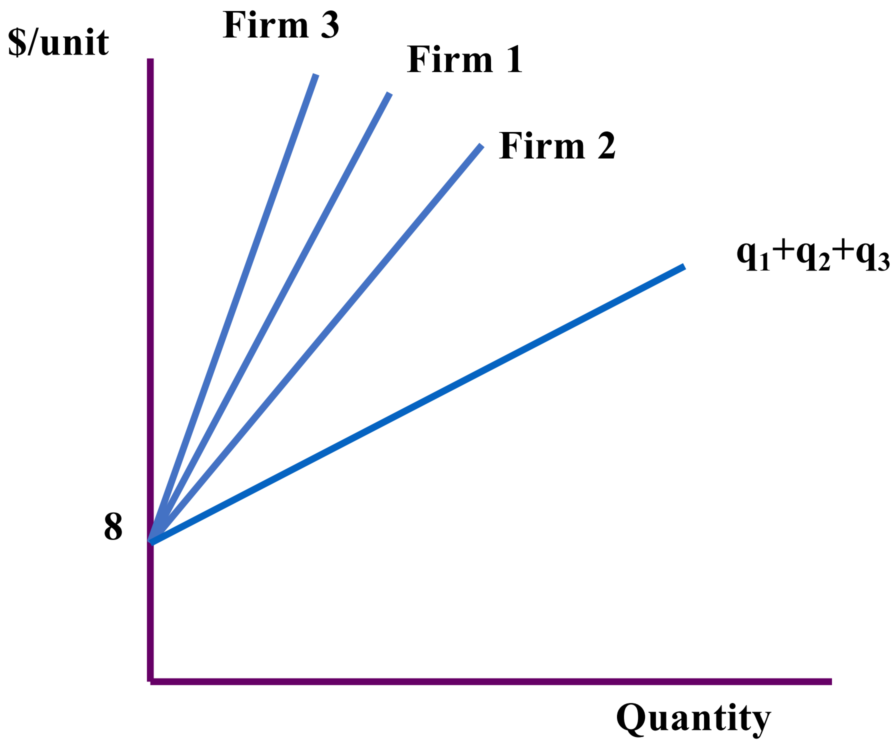
$$\text{Firm 2: } q_2 = \frac{MC}{2} - 4$$

$$\text{Firm 3: } q_3 = \frac{MC}{6} - \frac{4}{3}$$

Aggregate:

$$Q = q_1 + q_2 + q_3 = \frac{11\text{MC}}{12} - \frac{22}{3}$$

$$\Rightarrow MC = \frac{12Q}{11} + 8$$



Example: 80 firms

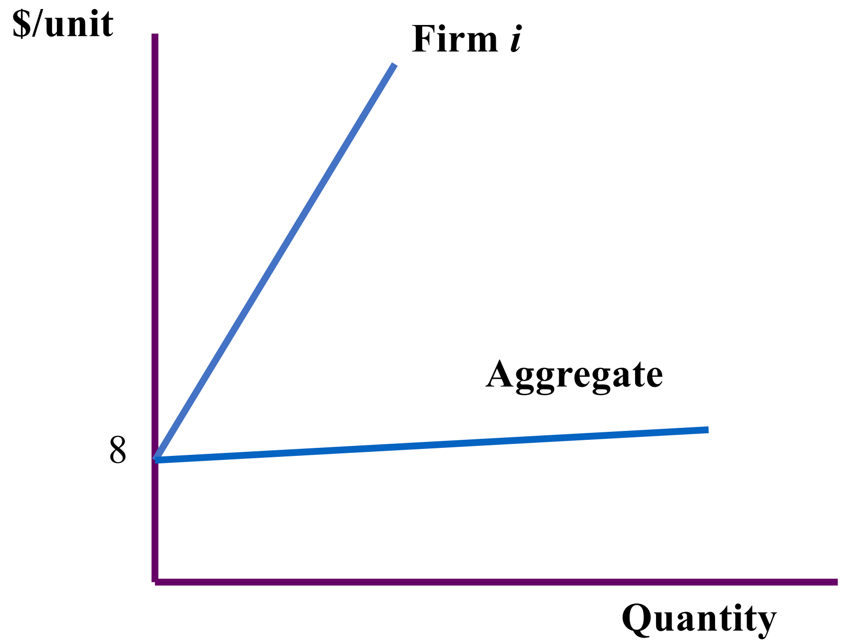
Each firm:

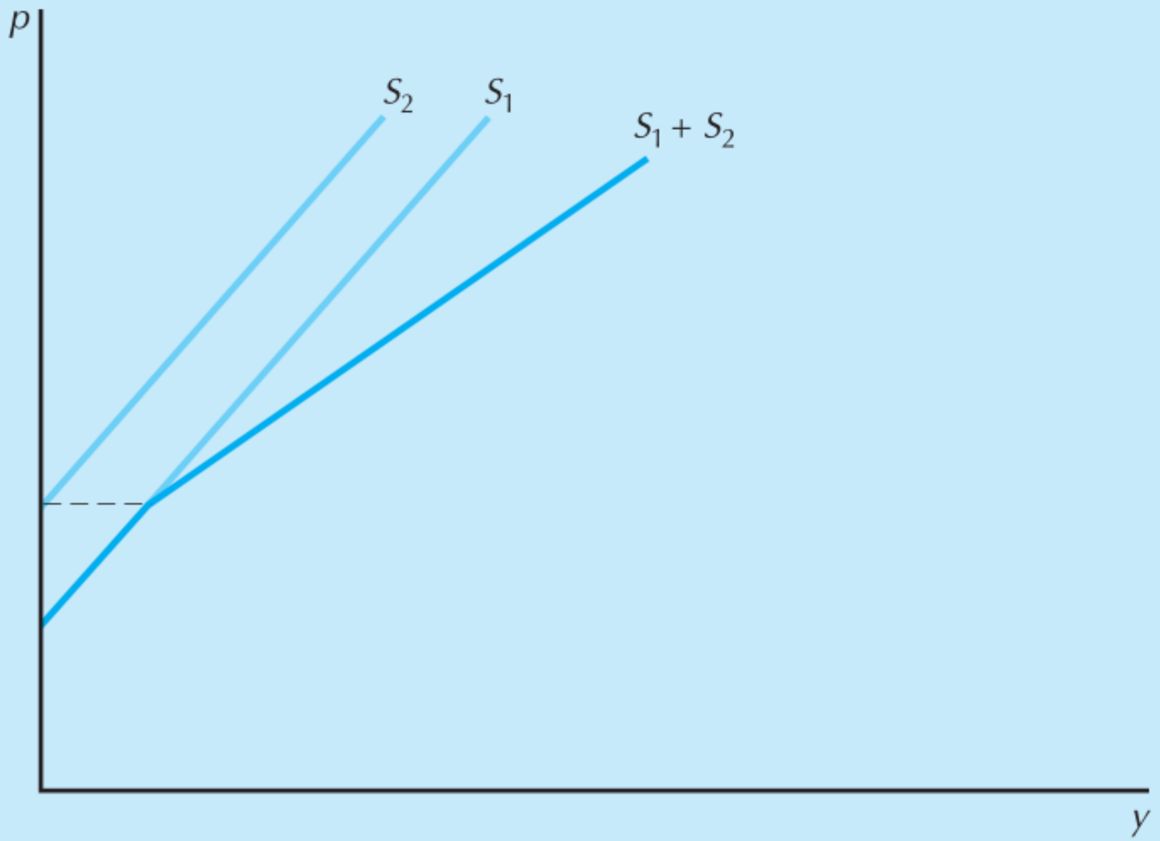
$$MC = 4q + 8 \Rightarrow q_i = \frac{MC}{4} - 2$$

Aggregate:

$$Q = 80q = 20MC - 160$$

$$\Rightarrow MC = \frac{Q}{20} + 8$$

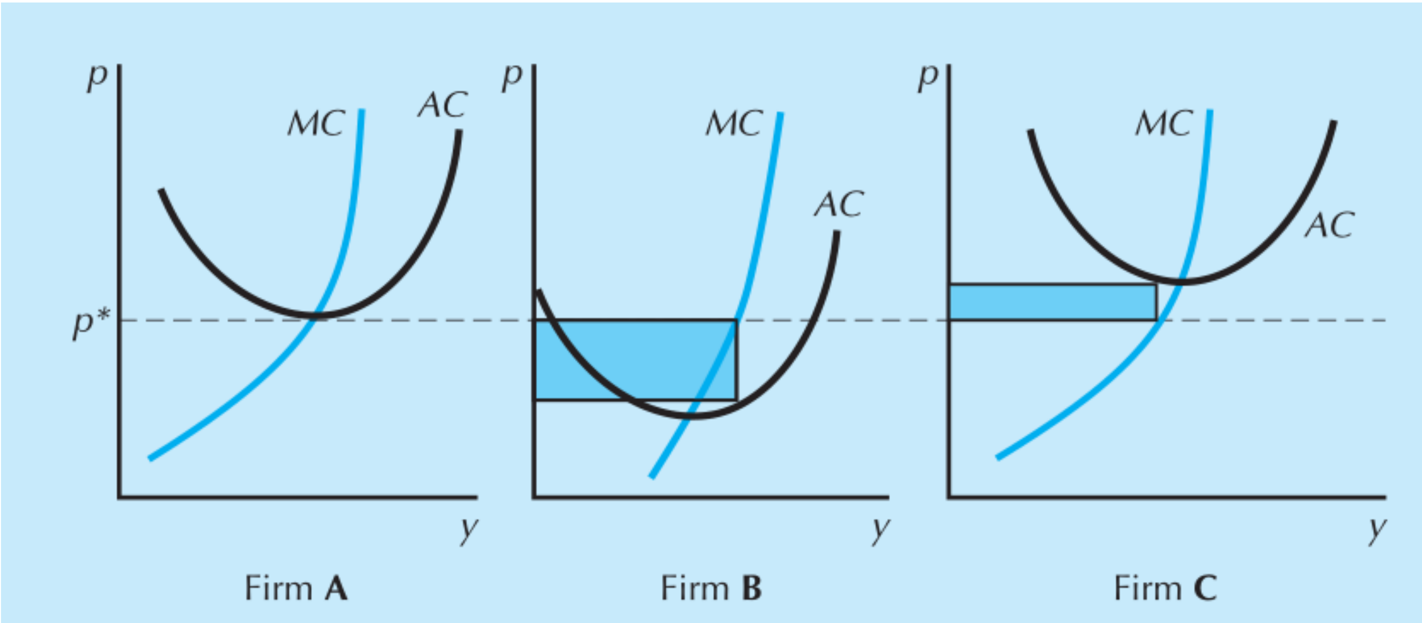




The industry supply curve. The industry supply curve $(S_1 + S_2)$ is the sum of the individual supply curves (S_1 and S_2).

Industry equilibrium in the short-run

- Find the intersection of the market supply curve with the market demand curve – gives equilibrium price p^*
- Given p^*, go back to the individual firm supply curves

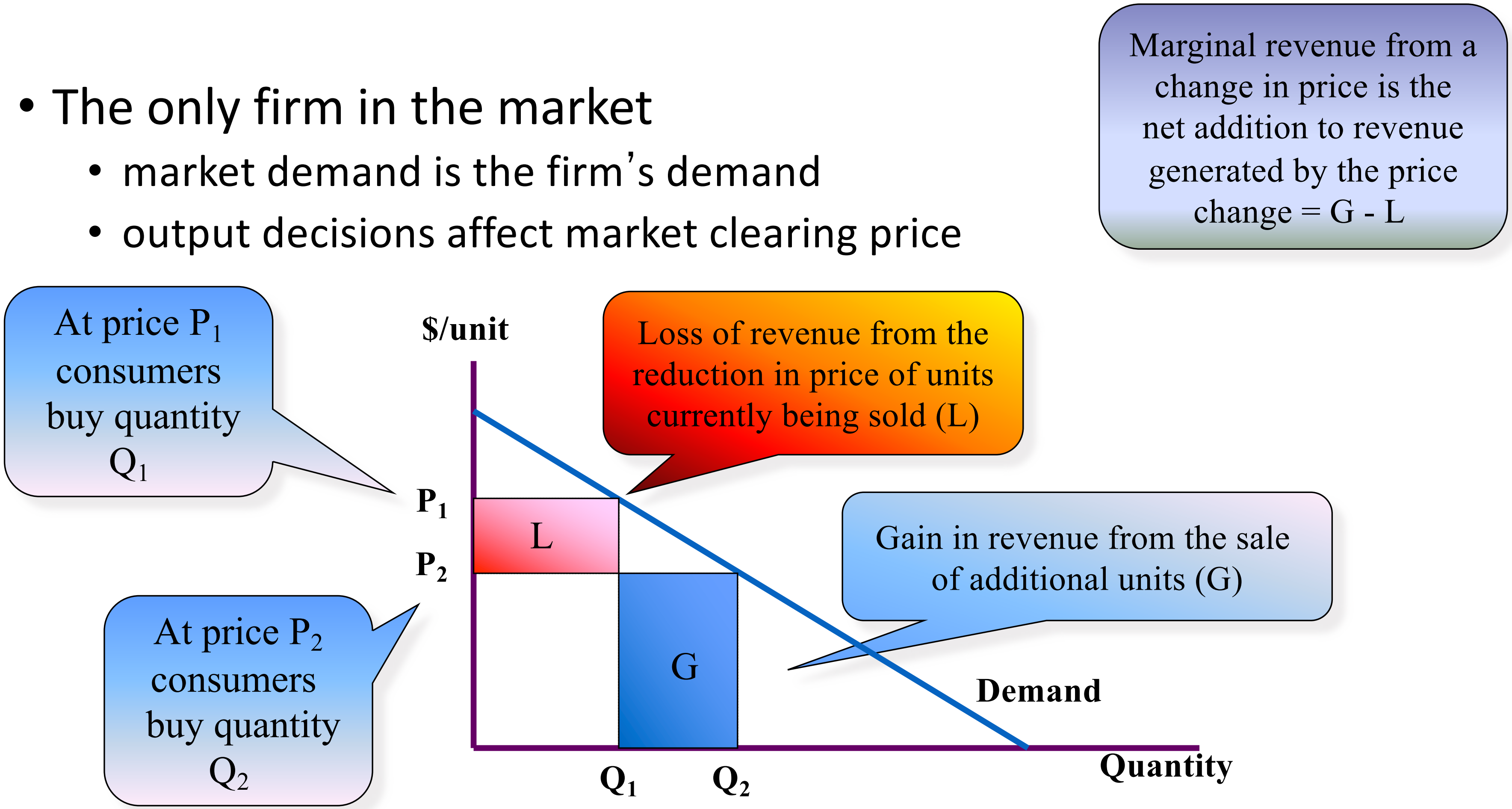


Industry equilibrium in the long-run

- In the long run, firms are able to adjust
- If firm is making losses, they can *exit*
- If firm is making profits, expect *entry* if there are no barriers to entry
- In a perfectly competitive market, profits = 0

Monopoly

- The only firm in the market
  - market demand is the firm's demand
  - output decisions affect market clearing price



Linear demand curve and monopoly

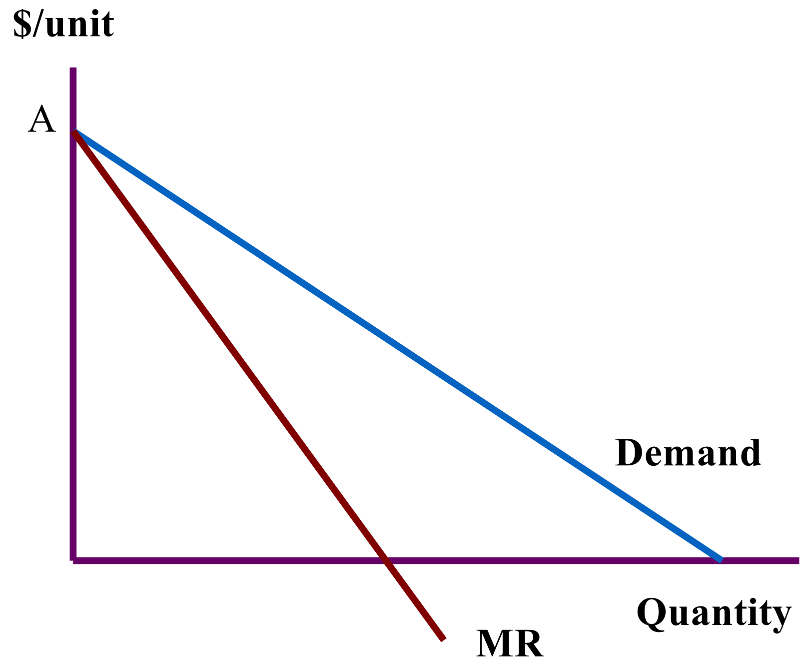
- Demand $P = A - BQ$
- Total Revenue

$TR = P \cdot Q = AQ - BQ^2$

- Marginal Revenue

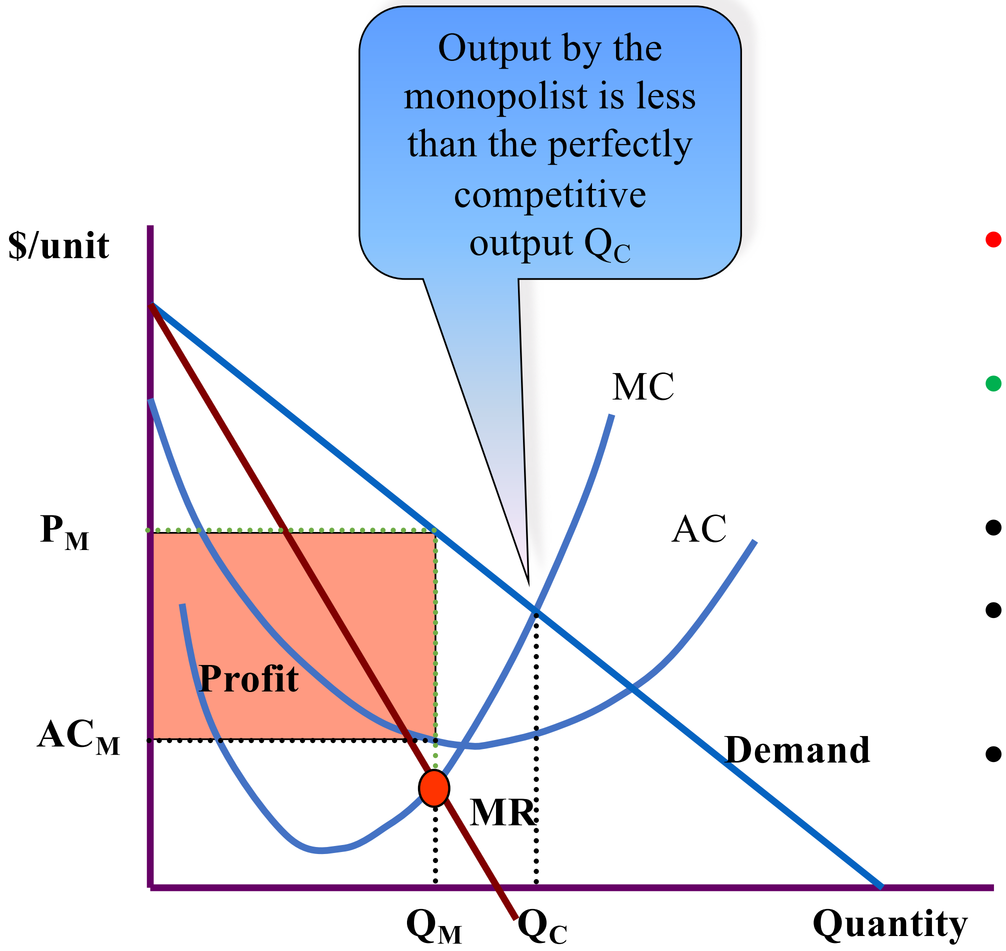
$MR = \frac{dTR}{dQ} = A - 2BQ$

- Marginal revenue curve is also linear with the same price intercept but twice the slope of the demand curve



Profit maximization

- The monopolist maximizes profit by equating marginal revenue with marginal cost
- This is a two-step process



- Step 1: Choose output where $MR = MC$, giving output Q_M
- Step 2: Identify the market clearing price P_M
- MR is less than price
- Price is greater than MC: loss of efficiency
- Price is greater than average cost
  - Positive economic profit
  - Long-run equilibrium: no entry

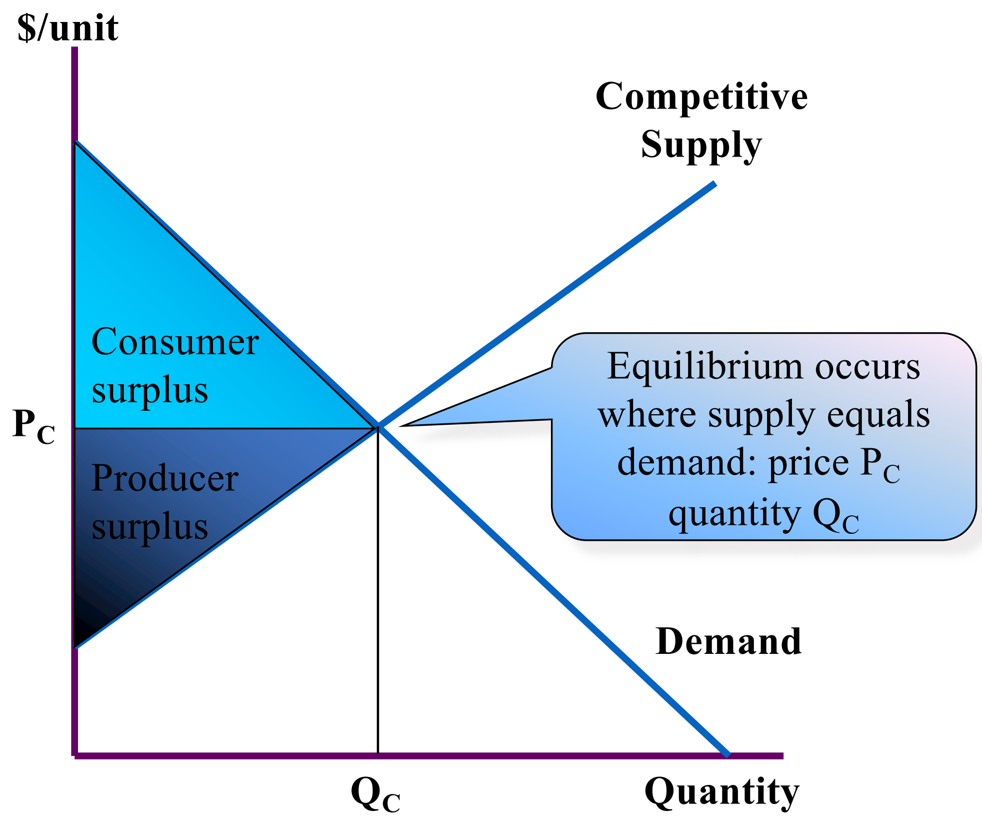
Efficiency and surplus

- Can we reallocate resources to make some individuals better off without making others worse off?
- Need a measure of well-being
  - ***Consumer surplus:*** difference between the maximum amount a consumer is willing to pay for a unit of a good and the amount actually paid for that unit
  - aggregate consumer surplus is the sum over all units consumed and all consumers

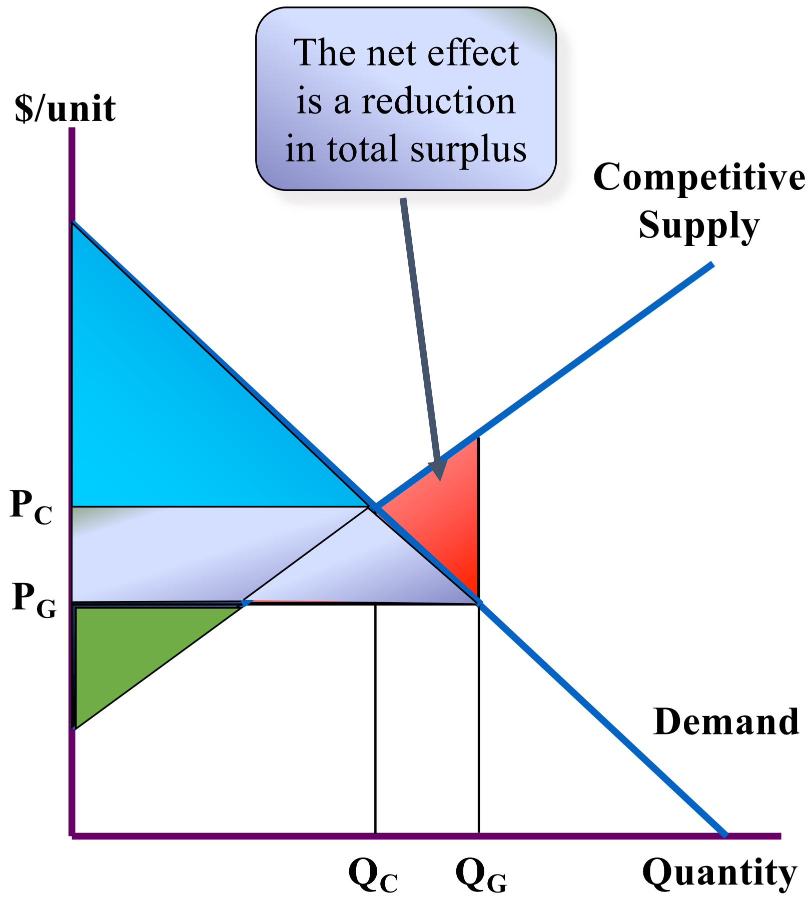
- ***Producer surplus:*** difference between the amount a producer receives from the sale of a unit and the amount that unit costs to produce
- aggregate producer surplus is the sum over all units produced and all producers
- ***total surplus*** = consumer surplus + producer surplus

Efficiency and surplus: illustration

- The demand curve measures the willingness to pay for each unit
  - Consumer surplus is the area between the demand curve and the equilibrium price
- The supply curve measures the marginal cost of each unit
  - Producer surplus is the area between the supply curve and the equilibrium price
- Total surplus is the sum of consumer surplus and producer surplus
- **The competitive equilibrium is efficient**

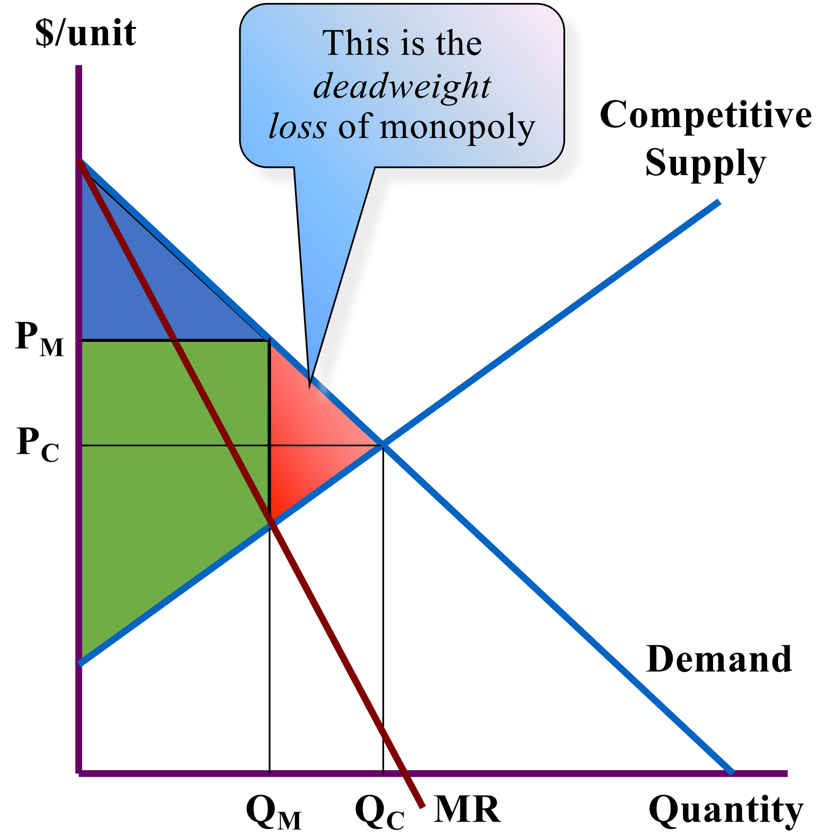


- Assume that a greater quantity Q_G is traded
- Price falls to P_G
- Producer surplus is now a positive part and a negative part
- Consumer surplus increases
- Part of this is a transfer from producers
- Part offsets the negative producer surplus



Deadweight loss of Monopoly

- The monopolist sets MR = MC to give output Q_M
- The market clearing price is P_M
- Consumer surplus
- Producer surplus
- The monopolist produces less surplus than the competitive industry.
- There are mutually beneficial trades that do not take place: between Q_M and Q_C



- Why can the monopolist not appropriate the deadweight loss?
  - Increasing output requires a reduction in price
  - *this assumes that the same price is charged to everyone.*
- The monopolist creates surplus
  - some goes to consumers
  - some appears as profit
- The monopolist bases her decisions purely on the surplus she gets, *not* on consumer surplus
- The monopolist undersupplies relative to the competitive outcome
- The primary problem: *the monopolist is large relative to the market*

A simple example

- Monopolist owns 2 units of a valuable good
- There are 50 000 potential buyers

Number of Buyers	Willingness To Pay
First 100	50 000
Next 40 000	15 000
Last 9 900	10 000

Both units will be sold at $50,000; no deadweight loss

Why not? Monopolist is *small* relative to the market

A simple example

- Monopolist owns 200 units of a valuable good
- There are 50 000 potential buyers

Number of Buyers	Willingness To Pay
First 100	50 000
Next 40 000	15 000
Last 9 900	10 000

100 units will be sold at $50,000; deadweight loss!

Is a monopoly necessarily bad?